

Retail Sales Analysis SQL Project

Project Overview

This project focuses on analyzing retail sales data using SQL. The dataset contains transactional sales records including customer details, product categories, quantities, and sales amounts. The goal is to clean the data, explore key metrics, and answer business-related questions to generate meaningful insights.

Dataset Description

The dataset (`sales_analysis`) includes the following columns:

- `sale_date` - Date of transaction
 - `sale_time` - Time of transaction
 - `customer_id` - Unique customer identifier
 - `gender` - Customer gender
 - `age` - Customer age
 - `category` - Product category (e.g., Clothing, Beauty, etc.)
 - `quantity` - Number of items purchased
 - `price_per_unit` - Price of each unit
 - `cogs` - Cost of goods sold
 - `total_sale` - Total transaction value
 - `transaction_id` - Unique transaction identifier
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Data Cleaning Process

- Checked for missing values in all key columns
- Removed rows containing NULL values to ensure data quality
- Disabled safe update mode to allow deletion of invalid records

```
DELETE FROM sales_analysis
WHERE sale_date IS NULL
   OR sale_time IS NULL
   OR customer_id IS NULL
   OR gender IS NULL
   OR age IS NULL
   OR category IS NULL
   OR quantity IS NULL
   OR price_per_unit IS NULL
   OR cogs IS NULL;
```

Exploratory Data Analysis (EDA)

Key exploration steps included:

- Counting total transactions
- Identifying unique customers
- Listing product categories

```
SELECT COUNT(*) FROM sales_analysis;  
SELECT COUNT(DISTINCT customer_id) FROM sales_analysis;  
SELECT DISTINCT category FROM sales_analysis;
```

Business Questions & Analysis

The following business questions were answered using SQL:

1. Retrieve all sales made on a specific date
2. Find clothing purchases with quantity ≥ 4 in Nov 2022
3. Calculate total sales and number of orders per category
4. Find average age of Beauty category customers
5. Identify transactions with total sales > 1000
6. Count transactions by gender and category
7. Determine best-selling month per year using window functions
8. Find top 5 customers by total sales
9. Count unique customers per category
10. Create time-based shifts (Morning, Afternoon, Evening) and count orders

Key Insights

- Clothing and Beauty categories showed strong customer engagement
- A small group of customers contributed to a large portion of total sales
- Sales activity is distributed across different time shifts (Morning, Afternoon, Evening)
- Seasonal trends can be identified using monthly aggregation

Tools & Technologies

- MySQL
 - SQL (DDL, DML, Aggregations, CTEs, Window Functions)
 - Data Cleaning & Transformation techniques
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How to Use

1. Import the dataset into MySQL
 2. Run the SQL script step by step
 3. Start with data cleaning section
 4. Proceed to EDA and business analysis queries
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Author

Created as part of a SQL learning project focused on data analysis and portfolio development.